3. Evolution :

(a) Theories of origin of life.

(b) Theories of evolution; Natural selection, role of mutations in evolution, evolutionary patterns, molecular drive, mimicry, variation, isolation and speciation.

(c) Evolution of horse, elephant and man using fossil data.

(d) Hardy-Weinberg Law.

(e) Continental drift and distribution of animals.

4. Systematics—Zoological nomenclature, international code, cladistics, molecular taxonomy and biodiversity.

5. Biochemistry :

(a) Structure and role of carbohydrates, fats, fatty acids and cholesterol, proteins and amino-acids, nucleic acids. Bioenergetics.

(b) Glycolysis and Kreb cycle, oxidation and reduction, oxidative phosphorylation, energy conservation and release, ATP cycle, cyclic AMP – its structure and role.

(c) Hormone classification (steroid and peptide hormones), biosynthesis and functions.

(d) Enzymes: types and mechanisms of action.

(e) Vitamins and co-enzymes

(f) Immunoglobulin and immunity.

6. Physiology (with special reference to mammals) :

(a) Composition and constituents of blood; blood groups and Rh factor in man, factors and mechanism of coagulation, iron metabolism, acid-base balance, thermo-regulation, anticoagulants.

(b) Haemoglobin : Composition, types and role in transport of oxygen and carbon dioxide.

(c) Digestion and absorption: Role of salivary glands, liver, pancreas and intestinal glands.

(d) Excretion : nephron and regulation of urine formation; osmo-regulation and excretory product

(e) Muscles : Types, mechanism of contraction of skeletal muscles, effects of exercise on muscles.

(f) Neuron : nerve impulse–its conduction and synaptic transmission, neurotransmitters.

(g) Vision, hearing and olfaction in man.

(h) Physiology of reproduction, puberty and menopause in human.

7. Developmental Biology :

(a) Gametogenesis; spermatogenesis, composition of semen, *in vitro* and *in vivo* capacitation of mammalian sperm, Oogenesis, totipotency; fertilization, morphogenesis and morphogen, blastogenesis, establishment of body axes formation, fate map, gestulation in frog and chick; genes in development in chick, homeotic genes, development of eye and heart, placenta in mammals.

(b) Cell lineage, cell-to cell interaction, Genetic and induced teratogenesis, role of thyroxine in control of metamorphosis in amphibia, paedogenesis and neoteny, cell death, aging.

(c) Developmental genes in man, *in vitro* fertilization and embryo transfer, cloning.

(d) Stem cells : Sources, types and their use in human welfare.

(e) Biogenetic law.
